



Lecture 7: Pandas, Matplotlib, EDA

Instructor: Jordan Schwartz

Slides adapted from Tony Liu

Logistics

- Nothing due yet... Have a good spring break!
- Please fill out the mid-semester feedback form sent out by Professor Sheth!
- Final project info will be posted on Ed soon! Keep an eye out!

What is module 3, Data Science, about? LAST WEEK

- **Lecture 6 - Jupyter/colab and Numpy**
 - What is data science
 - What are notebooks
 - Jupyter
 - How to open a jupyter notebook vs a colab notebook
 - Intro to numpy
 - Broadcasting and vectorization
 - If time- Intro to Matplotlib
 - Just a high level overview, go in depth next lecture

TODAY Week 7 - Pandas, Matplotlib, EDA

- What is pandas
 - What can pandas do
 - What is a dataframe
 - How to create a dataframe
- EDA example
 - Seaborn plotting



Pandas

What is Pandas?

- A powerful way to organize and manipulate data
- Real world jobs that use pandas (from indeed) →
- Honestly? It can be...
 - Pretty complex
 - Overwhelming at first
 - Not necessarily intuitive
- We're gonna jump right into the deep end

Data Scientist, Podcast Monetization

Spotify
New York, NY

\$94,584 - \$135,121 a year Paid parental leave Parental leave
401(k) Health insurance Meals & incidentals stipend

Bioinformatician A/B

University of Pennsylvania
Philadelphia, PA

\$57,500.00 - \$101,200.50 a year Full-time Tuition reimbursement
Health insurance Retirement plan Vision insurance

Data Scientist

BlueConic
Philadelphia, PA

\$90,000 - \$120,000 a year Full-time Health insurance
401(k) matching Vision insurance Dental insurance

Data Analyst C (Immune Health Institute)

University of Pennsylvania
Philadelphia, PA

\$61,000 - \$73,581 a year Full-time Tuition reimbursement
Health insurance Retirement plan Vision insurance

Entry-Level Data Science / Python Software Developer Position - US Citizen Only

AVNIK Defense Solutions, Inc.
Redstone Arsenal, AL

Pay information not provided Full-time

How familiar are we with sheets?

Let's look at some (relatively) small data and try to figure some stuff out!

CIS 1902 Lecture 7 – Small spotify dataset

Let's see how we can improve!

<https://colab.research.google.com/drive/10rPZHl3grEly8MXl0UWKg7Ev0emuKtFH#scrollTo=ll6qlxTMBbJD>

Thanks !

Do you have any questions?

youremail@freepik.com

+34 654 321 432

yourwebsite.com



CREDITS: This presentation template was created by **Slidesgo**, and includes icons, infographics & images by **Freepik**

Please keep this slide for attribution

Instructions for use

If you have a free account, in order to use this template, you must credit [Slidesgo](#) by keeping the [Thanks](#) slide. Please refer to the next slide to read the instructions for premium users.

As a Free user, you are allowed to:

- Modify this template.
- Use it for both personal and commercial projects.

You are not allowed to:

- Sublicense, sell or rent any of Slidesgo Content (or a modified version of Slidesgo Content).
- Distribute Slidesgo Content unless it has been expressly authorized by Slidesgo.
- Include Slidesgo Content in an online or offline database or file.
- Offer Slidesgo templates (or modified versions of Slidesgo templates) for download.
- Acquire the copyright of Slidesgo Content.

For more information about editing slides, please read our FAQs or visit our blog:

<https://slidesgo.com/faqs> and <https://slidesgo.com/slidesgo-school>